

altairsim — Altair 4K BASIC off a cassette

2026-10-01 · 43827d6

```
altairsim basic4k.toml

MEMORY SIZE? <return>
TERMINAL WIDTH? <return>
WANT SIN? Y

742 BYTES FREE

ALTAIR BASIC VERSION 3.1
[FOUR-K VERSION]

OK
```

Altair 4K BASIC 3.1, read off a period cassette image by the bootstrap that MITS shipped. Both are unmodified. The machine is the bare 1975 Altair: 4K of RAM, an 88-SIO for the Teletype, an 88-ACR for the recorder, and the sense switches at `0x80`.

The three lines in the `startup` list of `basic4k.toml` are the three things that an operator did in 1975:

1. Put the tape in the recorder and press PLAY (`MOUNT`).
2. Toggle in the bootstrap on the front panel (`LOAD`).
3. Run it from zero (`RUN 0`).

The bootstrap (`LDR4K31`) jumps into BASIC when the tape ends, so one command gets you to the prompt.

The tape loads in about one second. The real 300-baud cassette took 110 seconds. The difference is the default clock, which runs as fast as possible. Type `SET acr0:tape rate=real` before you run to play the tape at its real speed. BASIC gets the same result at either speed.

`basic4k.ini` builds the same machine with monitor commands: `altairsim -s basic4k.ini`.

The same tape in audio form

Most surviving cassettes are audio recordings, so this folder has the tape in audio form too. `4K BASIC Ver 3-1.wav` holds the program of the `.tap`, encoded in the audio format of the 88-ACR. `basic4k-wav.toml` mounts it and boots. The board decodes the audio in the same way that the real board did:

```
altairsim basic4k-wav.toml
```

`4K BASIC (Kansas City).wav` holds the same program, encoded in the Kansas City audio format. The 88-ACR cannot read that format, so the board refuses the tape when you mount it. The tapes chapter of the manual tells you why.

The files

File	What it is
<code>basic4k.toml</code>	The machine file. <code>base = "basic4k"</code> is the built-in hardware. This file puts the tape in the recorder and runs it.
<code>basic4k.ini</code>	The same machine, set up by monitor commands.
<code>basic4k-wav.toml</code>	The same machine, with the audio tape in the recorder.
<code>basic4k-wav.ini</code>	The same, set up by monitor commands.
<code>4K BASIC Ver 3-1.tap</code>	The cassette.
<code>4K BASIC Ver 3-1.wav</code>	The program, encoded in the audio format of the 88-ACR.
<code>4K BASIC (Kansas City).wav</code>	The program, encoded in the Kansas City audio format. The 88-ACR cannot read it.
<code>LDR4K31.HEX</code>	The bootstrap, assembled.

The bootstrap source, `LDR4K31.ASM` and `.PRN`, is in `tapes/4KBasic31/` in the repository. It is source, not product, so it is not in the package.